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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 25th February Morning Shift

JEE Main Question Paper with Answer Key

25 Feb 2021	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	22 questions
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Out of syllabus	18 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

The solubility of AgCN in a buffer solution of pH = 3 is x. The value of x is : [Assume : No cyano complex is formed; K_{sp}(AgCN) = 2.2×10^{-16} and K_a(HCN) = 6.2×10^{-10}]

Open solution

(A) 1.9×10^{-5} (B) 1.6×10^{-6} (C) 2.2×10^{-16} (D) 0.625×10^{-6}

Q2. MCQ (Single Correct Answer)

Which one of the following reactions will not form acetaldehyde?

Open solution

(A) $CH_3CH_2OH \xrightarrow[573K]{Cu}$

(B) $CH_3CH_2OH \xrightarrow{CrO_3-H_2SO_4}$

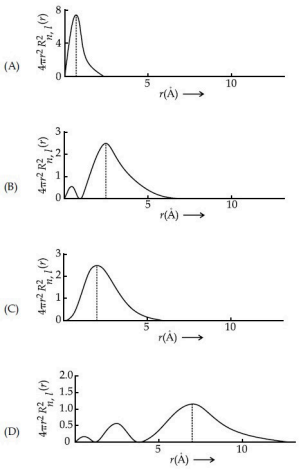
(C) $CH_2 = CH_2 + O_2 \xrightarrow[H_2O]{Pd(II)/Cu(II)}$

(D) $CH_3CN \xrightarrow[ii)\ H_2O]{i)\ DIBAL-H}$

Q3. MCQ (Single Correct Answer)

The plots of radial distribution functions for various orbitals of hydrogen atom against 'r' are given below :

Open solution



The correct plot for 3s orbital is :

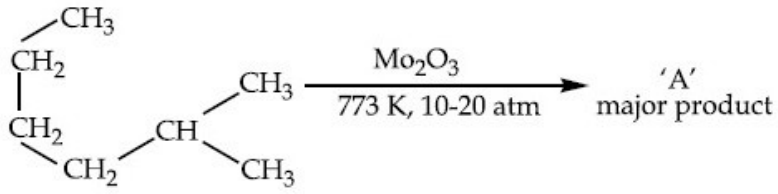
(A) (B) (B) (A) (C) (C) (D) (D)

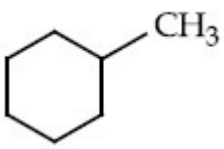
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Q4. MCQ (Single Correct Answer)

Open solution

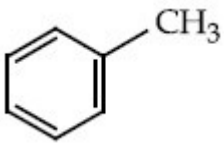
Identify A in the given chemical reaction.



(A) 

(B) 

(C) 

(D) 

Q5. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : CeO_2 can be used for oxidation of aldehydes and ketones.

Statement II : Aqueous solution of $EuSO_4$ is a strong reducing agent.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false.

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

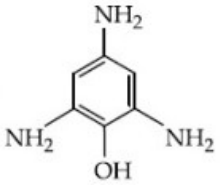
(D) Both Statement I and Statement II are true

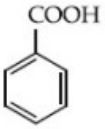
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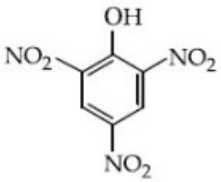
Q6. MCQ (Single Correct Answer)

Open solution

Compound(s) which will liberate carbon dioxide with sodium bicarbonate solution is/are :

A = 

B = 

C = 

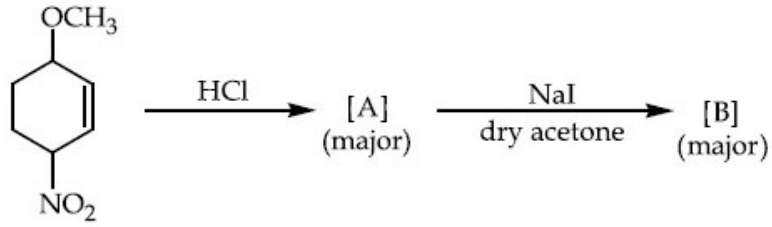
(A) C only (B) B only (C) B and C only (D) A and B only

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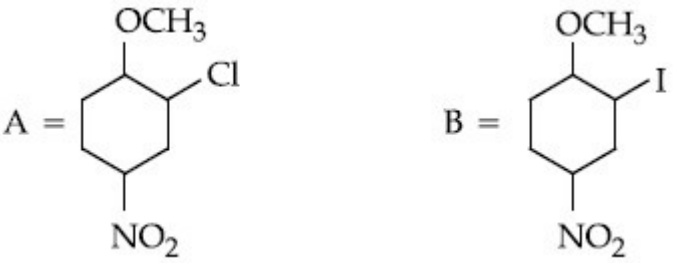
Q7. MCQ (Single Correct Answer)

Open solution

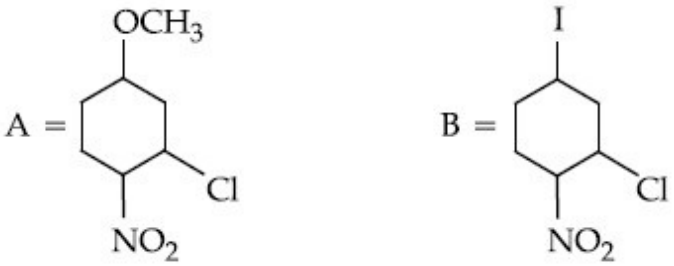
Identify A and B in the chemical reaction.



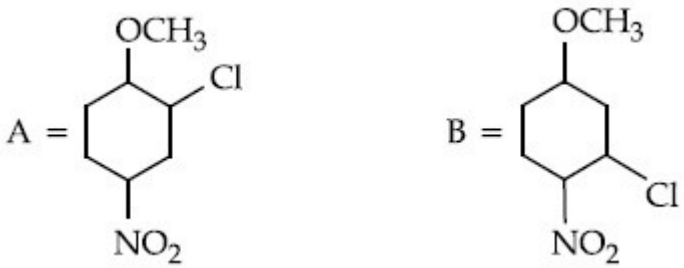
(A)



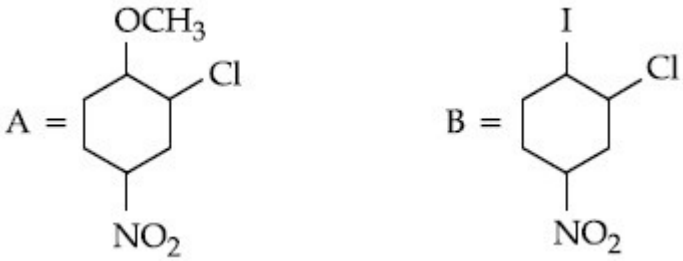
(B)



(C)



(D)



Q8. MCQ (Single Correct Answer)

Open solution

Which of the glycosidic linkage between galactose and glucose is present in lactose?

(A) C-1 of galactose and C-6 of glucose

(B) C-1 of glucose and C-6 of galactose

(C) C-1 of galactose and C-4 of glucose

(D) C-1 of glucose and C-4 of galactose

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Q9. MCQ (Single Correct Answer)

According to molecular orbital theory, the species among the following that does not exist is :

(A) O_2^{2-} (B) Be_2 (C) He_2^- (D) He_2^+

Open solution

Q10. MCQ (Single Correct Answer)

Complete combustion of 1.80 g of an oxygen containing compound ($C_xH_yO_z$) gave 2.64 g of CO_2 and 1.08 g of H_2O. The percentage of oxygen in the organic compound is :

(A) 51.63 (B) 50.33 (C) 63.53 (D) 53.33

Open solution

Q11. MCQ (Single Correct Answer)

The hybridization and magnetic nature of $[Mn(CN)_6]^{4-}$ and $[Fe(CN)_6]^{3-}$, respectively are :

(A) sp^3d^2 and diamagnetic (B) d^2sp^3 and paramagnetic

(C) sp^3d^2 and paramagnetic (D) d^2sp^3 and diamagnetic

Open solution

Q12. MCQ (Single Correct Answer)

The major product of the following chemical reaction is :

$$CH_3CH_2CN \xrightarrow[3)\ Pd/BaSO_4,\ H_2]{1)\ H_3O^+,\ \Delta \quad 2)\ SOCl_2} ?$$

(A) CH_3CH_2CHO (B) $CH_3CH_2CH_3$ (C) $CH_3CH_2CH_2OH$ (D) $(CH_3CH_2CO)_2O$

Open solution

Q13. MCQ (Single Correct Answer)

In which of the following pairs, the outer most electronic configuration will be the same?

(A) Cr^+ and Mn^{2+} (B) Ni^{2+} and Cu^+ (C) V^{2+} and Cr^+ (D) Fe^{2+} and Co^+

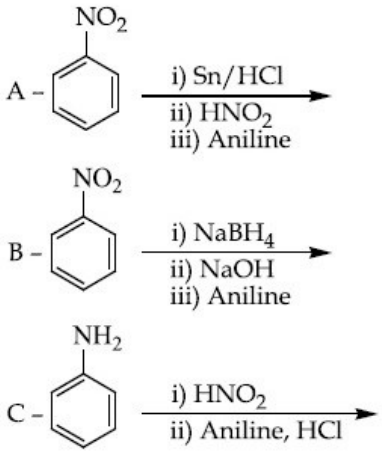
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Q14. MCQ (Single Correct Answer)

Open solution

Which of the following reaction/s will not give p-aminoazobenzene?



(A) C only **(B)** A only **(C)** A and B **(D)** B only

Q15. Numerical

Open solution

Using the provided information in the following paper chromatogram :

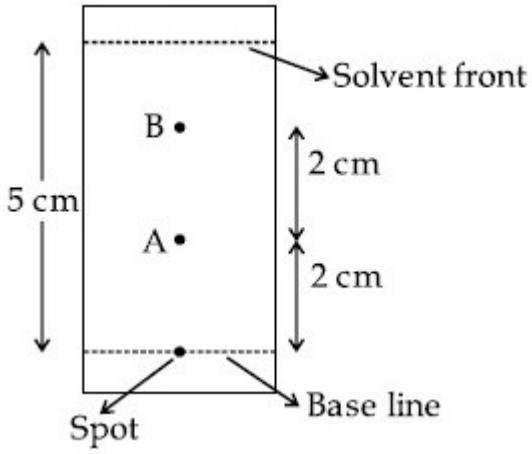


Fig : Paper chromatography for compounds A and B.

the calculated R_f value of A _____ $\times 10^{-1}$.

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Q16. Numerical

The reaction of cyanamide, $NH_2CN_{(s)}$ with oxygen was run in a bomb calorimeter and ΔU was found to be -742.24 kJ mol^{-1}. The magnitude of ΔH_{298} for the reaction

$$NH_2CN_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow N_{2(g)} + CO_{2(g)} + H_2O_{(l)}$$

is _____ kJ. (Rounded off to the nearest integer) [Assume ideal gases and R = 8.314 J mol^{-1} K^{-1}]

Open solution

Q17. Numerical

The ionization enthalpy of Na^+ formation from $Na_{(g)}$ is 495.8 kJ mol^{-1}, while the electron gain enthalpy of Br is -325.0 kJ mol^{-1}. Given the lattice enthalpy of NaBr is -728.4 kJ mol^{-1}. The energy for the formation of NaBr ionic solid is $(-)$ _____ $\times 10^{-1}$ kJ mol^{-1}.

Open solution

Q18. Numerical

0.4 g mixture of NaOH, Na_2CO_3 and some inert impurities was first titrated with $\frac{N}{10}$ HCl using phenolphthalein as an indicator, 17.5 mL of HCl was required at the end point. After this methyl orange was added and titrated. 1.5 mL of same HCl was required for the next end point. The weight percentage of Na_2CO_3 in the mixture is _____. (Rounded off to the nearest integer)

Open solution

Q19. Numerical

In basic medium CrO_4^{2-} oxidises $S_2O_3^{2-}$ to form SO_4^{2-} and itself changes into $Cr(OH)_4^-$. The volume of 0.154 M CrO_4^{2-} required to react with 40 mL of 0.25 M $S_2O_3^{2-}$ is _____ mL. (Rounded off to the nearest integer)

Open solution

Q20. Numerical

Consider the following chemical reaction.

$$CH \equiv CH \xrightarrow[2)\ CO,\ HCl,\ AlCl_3]{1)\ \text{Red hot Fe tube, 873 K}} \text{Product}$$

The number of sp^2 hybridized carbon atom(s) present in the product is _____.

Open solution

Q21. Numerical

1 molal aqueous solution of an electrolyte A_2B_3 is 60% ionised. The boiling point of the solution at 1 atm is _____ K. (Rounded off to the nearest integer)

[Given K_b for (H_2O) = 0.52 K kg mol^{-1}]

Open solution

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Q22. Numerical

Among the following, the number of halide(s) which is/are inert to hydrolysis is _____.

Open solution

(A) BF_3

(B) $SiCl_4$

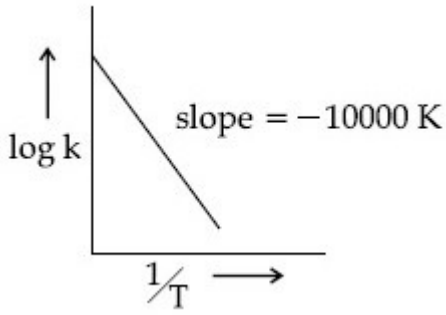
(C) PCl_5

(D) SF_6

Q23. Numerical

For the reaction, aA + bB $\rightarrow$ cC + dD, the plot of log k vs $\frac{1}{T}$ is given below :

Open solution



The temperature at which the rate constant of the reaction is 10^{-4} s^{-1} is _____ K. (Rounded off to the nearest integer)

[Given : The rate constant of the reaction is 10^{-5} s^{-1} at 500 K.]

Mathematics 22 questions

Q24. MCQ (Single Correct Answer)

When a missile is fired from a ship, the probability that it is intercepted is $\frac{1}{3}$ and the probability that the missile hits the target, given that it is not intercepted, is $\frac{3}{4}$. If three missiles are fired independently from the ship, then the probability that all three hit the target, is :

Open solution

(A) $\frac{3}{4}$ (B) $\frac{3}{8}$ (C) $\frac{1}{27}$ (D) $\frac{1}{8}$

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Q25. MCQ (Single Correct Answer)

The equation of the line through the point (0, 1, 2) and perpendicular to the line

$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{-2}$ is :

(A) $\frac{x}{3} = \frac{y-1}{-4} = \frac{z-2}{3}$

(B) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{-3}$

(C) $\frac{x}{-3} = \frac{y-1}{4} = \frac{z-2}{3}$

(D) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{3}$

Open solution

Q26. MCQ (Single Correct Answer)

The value of $\int_{-1}^{1} x^2 e^{[x^3]} dx$, where [t] denotes the greatest integer $\leq$ t, is :

(A) $\frac{e+1}{3}$

(B) $\frac{e-1}{3e}$

(C) $\frac{1}{3e}$

(D) $\frac{e+1}{3e}$

Open solution

Q27. MCQ (Single Correct Answer)

The integer 'k', for which the inequality $x^2 - 2(3k-1)x + 8k^2 - 7 > 0$ is valid for every x in R, is :

(A) 4

(B) 2

(C) 3

(D) 0

Open solution

Q28. MCQ (Single Correct Answer)

The total number of positive integral solutions (x, y, z) such that xyz = 24 is :

(A) 36

(B) 24

(C) 45

(D) 30

Open solution

Q29. MCQ (Single Correct Answer)

If a curve passes through the origin and the slope of the tangent to it at any point (x, y) is $\frac{x^2-4x+y+8}{x-2}$, then this curve also passes through the point :

(A) (4, 4)

(B) (5, 5)

(C) (5, 4)

(D) (4, 5)

Open solution

Q30. MCQ (Single Correct Answer)

Let f, g : N $\rightarrow$ N such that f(n + 1) = f(n) + f(1) $\forall$ n$\in$N and g be any arbitrary function. Which of the following statements is NOT true?

(A) If g is onto, then fog is one-one

(B) f is one-one

(C) If f is onto, then f(n) = n $\forall$n$\in$N

(D) If fog is one-one, then g is one-one

Open solution

Q31. MCQ (Single Correct Answer)

Let the lines $(2-i)z = (2+i)\bar{z}$ and $(2+i)z + (i-2)\bar{z} - 4i = 0$, (here $i^2 = -1$) be normal to a circle C. If the line $iz + \bar{z} + 1 + i = 0$ is tangent to this circle C, then its radius is :

(A) $\frac{3}{2\sqrt{2}}$

(B) $3\sqrt{2}$

(C) $\frac{1}{2\sqrt{2}}$

(D) $\frac{3}{\sqrt{2}}$

Open solution

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Q32. MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left(1+\frac{1+\frac{1}{2}+.....+\frac{1}{n}}{n^2}\right)^n$ is equal to :

(A) $\frac{1}{2}$ (B) 1 (C) 0 (D) $\frac{1}{e}$

Open solution

Q33. MCQ (Single Correct Answer)

Let α be the angle between the lines whose direction cosines satisfy the equations l + m − n = 0 and $l^2 + m^2 - n^2 = 0$. Then the value of $\sin^4\alpha + \cos^4\alpha$ is :

(A) $\frac{3}{8}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{5}{8}$

Open solution

Q34. MCQ (Single Correct Answer)

The value of the integral
$\int \frac{\sin\theta.\sin 2\theta(\sin^6\theta+\sin^4\theta+\sin^2\theta)\sqrt{2\sin^4\theta+3\sin^2\theta+6}}{1-\cos 2\theta}\,d\theta$ is :

(A) $\frac{1}{18}\left[9-2\cos^6\theta-3\cos^4\theta-6\cos^2\theta\right]^{\frac{3}{2}}+c$

(B) $\frac{1}{18}\left[11-18\sin^2\theta+9\sin^4\theta-2\sin^6\theta\right]^{\frac{3}{2}}+c$

(C) $\frac{1}{18}\left[11-18\cos^2\theta+9\cos^4\theta-2\cos^6\theta\right]^{\frac{3}{2}}+c$

(D) $\frac{1}{18}\left[9-2\sin^6\theta-3\sin^4\theta-6\sin^2\theta\right]^{\frac{3}{2}}+c$

Open solution

Q35. MCQ (Single Correct Answer)

The coefficients a, b and c of the quadratic equation, $ax^2 + bx + c = 0$ are obtained by throwing a dice three times. The probability that this equation has equal roots is :

(A) $\frac{1}{72}$ (B) $\frac{5}{216}$ (C) $\frac{1}{36}$ (D) $\frac{1}{54}$

Open solution

Q36. MCQ (Single Correct Answer)

The image of the point (3, 5) in the line x − y + 1 = 0, lies on :

(A) $(x-4)^2 + (y-4)^2 = 8$ (B) $(x-4)^2 + (y+2)^2 = 16$

(C) $(x-2)^2 + (y-2)^2 = 12$ (D) $(x-2)^2 + (y-4)^2 = 4$

Open solution

Q37. Numerical

The locus of the point of intersection of the lines $\left(\sqrt{3}\right)kx + ky - 4\sqrt{3} = 0$ and $\sqrt{3}x - y - 4\left(\sqrt{3}\right)k = 0$ is a conic, whose eccentricity is _____.

Open solution

Q38. Numerical

The graphs of sine and cosine functions, intersect each other at a number of points and between two consecutive points of intersection, the two graphs enclose the same area A. Then A^4 is equal to _____.

Open solution

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Q39. Numerical

The total number of numbers, lying between 100 and 1000 that can be formed with the digits 1, 2, 3, 4, 5, if the repetition of digits is not allowed and numbers are divisible by either 3 or 5, is _____.

Open solution

Q40. Numerical

If $A = \begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and $(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then $13(a^2 + b^2)$ is equal to

Open solution

Q41. Numerical

Let $A_1, A_2, A_3, \ldots\ldots$ be squares such that for each $n \geq 1$, the length of the side of A_n equals the length of diagonal of A_{n+1}. If the length of the side of A_1 is 12 cm, then the smallest value of n for which area of A_n is less than one, is _____.

Open solution

Q42. Numerical

If the system of equations

$kx + y + 2z = 1$

$3x - y - 2z = 2$

$-2x - 2y - 4z = 3$

has infinitely many solutions, then k is equal to _____.

Open solution

Q43. Numerical

The number of points, at which the function
$f(x) = |2x + 1| - 3|x + 2| + |x^2 + x - 2|$, $x \in R$ is not differentiable, is _____.

Open solution

Q44. Numerical

Let $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = \vec{c} \times \vec{a}$ and $\vec{r} \cdot \vec{b} = 0$, then $\vec{r} \cdot \vec{a}$ is equal to _____.

Open solution

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Q45. Numerical

Let f(x) be a polynomial of degree 6 in x, in which the coefficient of x^6 is unity and it has extrema at x = −1 and x = 1. If $\lim_{x\to 0}\frac{f(x)}{x^3}=1$, then $5.f(2)$ is equal to _____.

Open solution

Physics 27 questions

Q46. MCQ (Single Correct Answer)

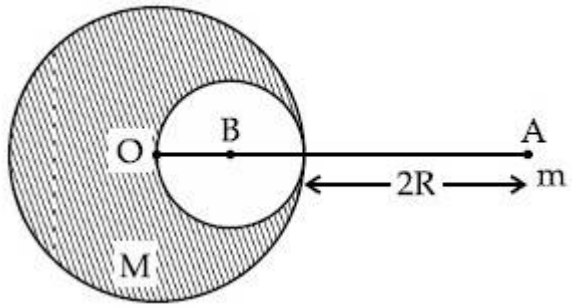
A proton, a deuteron and an α particle are moving with same momentum in a uniform magnetic field. The ratio of magnetic forces acting on them is _____ and their speed is _____, in the ratio.

(A) 2 : 1 : 1 and 4 : 2 : 1
(B) 4 : 2 : 1 and 2 : 1 : 1
(C) 1 : 2 : 4 and 2 : 1 : 1
(D) 1 : 2 : 4 and 1 : 1 : 2

Open solution

Q47. MCQ (Single Correct Answer)

A solid sphere of radius R gravitationally attracts a particle placed at 3R from its centre with a force F_1. Now a spherical cavity of radius $\left(\frac{R}{2}\right)$ is made in the sphere (as shown in figure) and the force becomes F_2. The value of $F_1 : F_2$ is



(A) 36 : 25
(B) 41 : 50
(C) 50 : 41
(D) 25 : 36

Open solution

Q48. MCQ (Single Correct Answer)

An α particle and a proton are accelerated from rest by a potential difference of 200V. After this, their de Broglie wavelengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$ is :

(A) 8
(B) 2.8
(C) 7.8
(D) 3.8

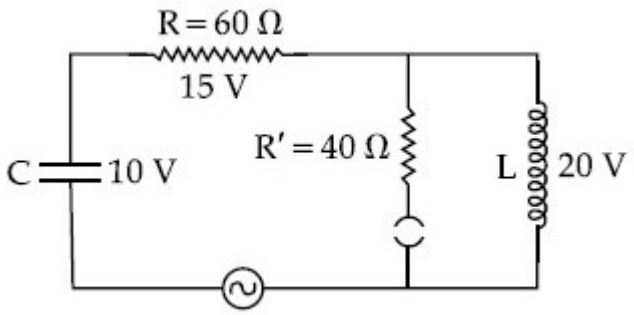
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Q49. MCQ (Single Correct Answer)

The angular frequency of alternating current in a L-C-R circuit is 100 rad/s. The components connected are shown in the figure. Find the value of inductance of the coil and capacity of condenser.

Open solution



(A) 0.8 H and 150 μF

(B) 0.8 H and 250 μF

(C) 1.33 H and 150 μF

(D) 1.33 H and 250 μF

Q50. MCQ (Single Correct Answer)

Magnetic fields at two points on the axis of a circular coil at a distance of 0.05 m and 0.2 m from the centre are in the ratio 8 : 1. The radius of coil is _____.

Open solution

(A) 1.0 m

(B) 0.15 m

(C) 0.2 m

(D) 0.1 m

Q51. MCQ (Single Correct Answer)

Match List - I with List - II :

Open solution

	List I		List II
(a)	h (Planck's constant)	(i)	$[MLT^{-1}]$
(b)	E (kinetic energy)	(ii)	$[ML^2T^{-1}]$
(c)	V (electric potential)	(iii)	$[ML^2T^{-2}]$
(d)	P (linear momentum)	(iv)	$[ML^2I^{-1}T^{-3}]$

Choose the correct answer from the options given below :

(A) (a) → (ii), (b) → (iii), (c) → (iv), (d) → (i)

(B) (a) → (i), (b) → (ii), (c) → (iv), (d) → (iii)

(C) (a) → (iii), (b) → (ii), (c) → (iv), (d) → (i)

(D) (a) → (iii), (b) → (iv), (c) → (ii), (d) → (i)

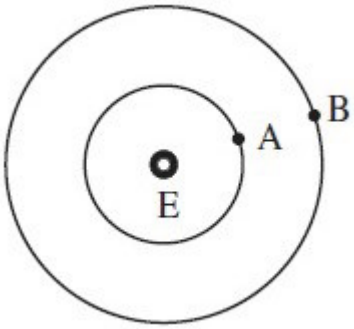
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Q52. MCQ (Single Correct Answer)

Open solution

Two satellites A and B of masses 200 kg and 400 kg are revolving round the earth at height of 600 km and 1600 km respectively.

If T_A and T_B are the time periods of A and B respectively then the value of $T_B - T_A$:



[Given : radius of earth = 6400 km, mass of earth = 6×10^{24} kg]

(A) 1.33×10^3 s (B) 4.24×10^2 s (C) 3.33×10^2 s (D) 4.24×10^3 s

Q53. MCQ (Single Correct Answer)

Open solution

Two coherent light sources having intensity in the ratio 2x produce an interference pattern. The ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be :

(A) $\frac{2\sqrt{2x}}{2x+1}$ (B) $\frac{2\sqrt{2x}}{x+1}$ (C) $\frac{\sqrt{2x}}{x+1}$ (D) $\frac{\sqrt{2x}}{2x+1}$

Q54. MCQ (Single Correct Answer)

Open solution

An engine of a train, moving with uniform acceleration, passes the signal-post with velocity u and the last compartment with velocity v. The velocity with which middle point of the train passes the signal post is :

(A) $\frac{u+v}{2}$ (B) $\sqrt{\frac{v^2-u^2}{2}}$

(C) $\frac{v-u}{2}$ (D) $\sqrt{\frac{v^2+u^2}{2}}$

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Q55. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : When a rod lying freely is heated, no thermal stress is developed in it.

Reason R : On heating, the length of the rod increases.

In the light of the above statements, choose the correct answer from the options given below :

(A) A is true but R is false

(B) A is false but R is true

(C) Both A and B are true but R is NOT the correct explanation of A

(D) Both A and R are true and R is the correct explanation of A

Q56. MCQ (Single Correct Answer)

Open solution

The pitch of the screw gauge is 1 mm and there are 100 divisions on the circular scale. When nothing is put in between the jaws, the zero of the circular scale lies 8 divisions below the reference line. When a wire is placed between the jaws, the first linear scale division is clearly visible while 72nd division on circular scale coincides with the reference line. The radius of the wire is :

(A) 1.80 mm (B) 0.90 mm (C) 0.82 mm (D) 1.64 mm

Q57. MCQ (Single Correct Answer)

Open solution

If the time period of a two meter long simple pendulum is 2s, the acceleration due to gravity at the place where pendulum is executing S.H.M. is :

(A) 16 m/s^2 (B) $2\pi^2$ ms^{-2} (C) π^2 ms^{-2} (D) 9.8 ms^{-2}

Q58. MCQ (Single Correct Answer)

Open solution

A diatomic gas, having $C_p = \frac{7}{2}R$ and $C_v = \frac{5}{2}R$, is heated at constant pressure. The ratio dU : dQ : dW :

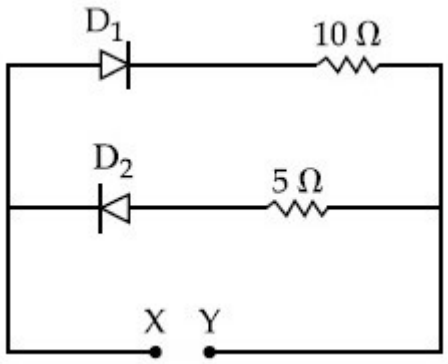
(A) 5 : 7 : 3 (B) 3 : 7 : 2 (C) 5 : 7 : 2 (D) 3 : 5 : 2

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Q59. MCQ (Single Correct Answer)

A 5V battery is connected across the points X and Y. Assume D_1 and D_2 to be normal silicon diodes. Find the current supplied by the battery if the +ve terminal of the battery is connected to point X.

Open solution



(A) $\sim$ 0.5 A (B) $\sim$ 1.5 A (C) $\sim$ 0.43 A (D) $\sim$ 0.86 A

Q60. MCQ (Single Correct Answer)

A student is performing the experiment of resonance column. The diameter of the column tube is 6 cm. The frequency of the tuning fork is 504 Hz. Speed of the sound at the given temperature is 336 m/s. The zero of the metre scale coincides with the top end of the resonance column tube. The reading of the water level in the column when the first resonance occurs is :

Open solution

(A) 13 cm (B) 18.4 cm (C) 16.6 cm (D) 14.8 cm

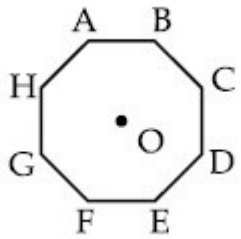
Q61. MCQ (Single Correct Answer)

In an octagon ABCDEFGH of equal side, what is the sum of

$\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} + \overrightarrow{AG} + \overrightarrow{AH}$,

Open solution

if, $\overrightarrow{AO} = 2\hat{i} + 3\hat{j} - 4\hat{k}$



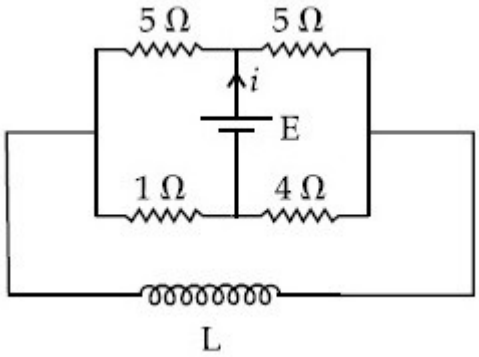
(A) $-16\hat{i} - 24\hat{j} + 32\hat{k}$ (B) $16\hat{i} + 24\hat{j} - 32\hat{k}$

(C) $16\hat{i} + 24\hat{j} + 32\hat{k}$ (D) $16\hat{i} - 24\hat{j} + 32\hat{k}$

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Q62. MCQ (Single Correct Answer)

The current (i) at time t = 0 and t = ∞ respectively for the given circuit is :



Open solution

(A) $\frac{18E}{55}, \frac{5E}{18}$

(B) $\frac{5E}{18}, \frac{18E}{55}$

(C) $\frac{5E}{18}, \frac{10E}{33}$

(D) $\frac{10E}{33}, \frac{5E}{18}$

Q63. MCQ (Single Correct Answer)

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : The escape velocities of planet A and B are same. But A and B are of unequal mass.

Reason R : The product of their mass and radius must be same. $M_1R_1 = M_2R_2$

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are correct and R is the correct explanation of A

(B) Both A and R are correct but R is NOT the correct explanation of A

(C) A is correct but R is not correct

(D) A is not correct but R is correct

Open solution

Q64. Numerical

A small bob tied at one end of a thin string of length 1 m is describing a vertical circle so that the maximum and minimum tension in the string are in the ratio 5 : 1. The velocity of the bob at the highest position is _____ m/s. (Take g = 10 m/s^2)

Open solution

Q65. Numerical

A monoatomic gas of mass 4.0 u is kept in an insulated container. Container is moving with velocity 30 m/s. If container is suddenly stopped then change in temperature of the gas (R = gas constant) is $\frac{x}{3R}$. Value of x is _____.

Open solution

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Q66. Numerical

The same size images are formed by a convex lens when the object is placed at 20 cm or at 10 cm from the lens. The focal length of convex lens is _____ cm.

Open solution

Q67. Numerical

A coil of inductance 2 H having negligible resistance is connected to a source of supply whose voltage is given by V = 3t volt. (where t is in second). If the voltage is applied when t = 0, then the energy stored in the coil after 4 s is _____J.

Open solution

Q68. Numerical

The potential energy (U) of a diatomic molecule is a function dependent on r (interatomic distance) as

$U = \frac{\alpha}{r^{10}} - \frac{\beta}{r^5} - 3$

where, α and β are positive constants. The equilibrium distance between two atoms will be $\left(\frac{2\alpha}{\beta}\right)^{\frac{a}{b}}$, where a = _____.

Open solution

Q69. Numerical

The electric field in a region is given by $\vec{E} = \left(\frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0\hat{j}\right)\frac{N}{C}$. The ratio of flux of reported field through the rectangular surface of area 0.2 m^2 (parallel to y $-$ z plane) to that of the surface of area 0.3 m^2 (parallel to x $-$ z plane) is a : b, where a = _____ [Here $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x, y and z-axes respectively.]

Open solution

Q70. Numerical

In a certain thermodynamical process, the pressure of a gas depends on its volume as kV^3. The work done when the temperature changes from 100°C to 300°C will be _____ nR, where n denotes number of moles of a gas.

Open solution

Q71. Numerical

512 identical drops of mercury are charged to a potential of 2V each. The drops are joined to form a single drop. The potential of this drop is _____ V.

Open solution

Q72. Numerical

A transmitting station releases waves of wavelength 960 m. A capacitor of 2.56 μF is used in the resonant circuit. The self inductance of coil necessary for resonance is _____ $\times 10^{-8}$ H.

Open solution

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Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q73. MCQ (Single Correct Answer) · **Chemistry**

Which of the following equation depicts the oxidizing nature of H_2O_2?

(A) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$

(B) $KIO_4 + H_2O_2 \rightarrow KIO_3 + H_2O + O_2$

(C) $2I^- + H_2O_2 + 2H^+ \rightarrow I_2 + 2H_2O$

(D) $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : An allotrope of oxygen is an important intermediate in the formation of reducing smog.

Statement II : Gases such as oxides of nitrogen and sulphur present in troposphere contribute to the formation of photochemical smog.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

The correct statement about B_2H_6 is :

(A) Its fragment, BH_3, behaves as a Lewis base.

(B) All B–H–B angles are of 120°.

(C) The two B–H–B bonds are not of same length.

(D) Terminal B–H bonds have less p-character when compared to bridging bonds.

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

In Freundlich adsorption isotherm at moderate pressure, the extent of adsorption $\left(\frac{x}{m}\right)$ is directly proportional to P^x. The value of x is :

(A) ∞ (B) zero (C) 1 (D) $\frac{1}{n}$

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Ellingham diagram is a graphical representation of :

(A) ΔG vs P

(B) ΔG vs T

(C) $(\Delta G - T\Delta S)$ vs T

(D) ΔH vs T

Open solution

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Q78. MCQ (Single Correct Answer) · **Chemistry**

Which statement is correct?

(A) Buna-N is a natural polymer.

(B) Synthesis of Buna-S needs nascent oxygen.

(C) Buna-S is a synthetic and linear thermosetting polymer.

(D) Neoprene is an addition copolymer used in plastic bucket manufacturing.

Open solution

Q79. Numerical · **Chemistry**

A car tyre is filled with nitrogen gas at 35 psi at 27°C. It will burst if pressure exceeds 40 psi. The temperature in °C at which the car tyre will burst is _____. (Rounded off to the nearest integer)

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in :

(A) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$

(B) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

(C) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(D) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If $0 < \theta, \phi < \frac{\pi}{2}, x = \sum_{n=0}^{\infty} \cos^{2n}\theta, y = \sum_{n=0}^{\infty} \sin^{2n}\phi$ and $z = \sum_{n=0}^{\infty} \cos^{2n}\theta . \sin^{2n}\phi$ then :

(A) xy − z = (x + y)z

(B) xyz = 4

(C) xy + z = (x + y)z

(D) xy + yz + zx = z

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The statement A $\rightarrow$ (B $\rightarrow$ A) is equivalent to :

(A) A $\rightarrow$ (A $\longleftrightarrow$ B)

(B) A $\rightarrow$ (A $\vee$ B)

(C) A $\rightarrow$ (A $\wedge$ B)

(D) A $\rightarrow$ (A $\rightarrow$ B)

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

A tangent is drawn to the parabola $y^2 = 6x$ which is perpendicular to the line 2x + y = 1. Which of the following points does NOT lie on it?

(A) (0, 3)

(B) (−6, 0)

(C) (4, 5)

(D) (5, 4)

Open solution

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Q84. MCQ (Single Correct Answer) · **Mathematics**

A man is observing, from the top of a tower, a boat speeding towards the lower from a certain point A, with uniform speed. At that point, angle of depression of the boat with the man's eye is 30° (Ignore man's height). After sailing for 20 seconds, towards the base of the tower (which is at the level of water), the boat has reached a point B, where the angle of depression is 45°. Then the time taken (in seconds) by the boat from B to reach the base of the tower is :

Open solution

(A) $10(\sqrt{3}+1)$ (B) $10(\sqrt{3}-1)$ (C) 10 (D) $10\sqrt{3}$

Q85. MCQ (Single Correct Answer) · **Mathematics**

If the curves, $\frac{x^2}{a}+\frac{y^2}{b}=1$ and $\frac{x^2}{c}+\frac{y^2}{d}=1$ intersect each other at an angle of 90°, then which of the following relations is TRUE?

Open solution

(A) a − c = b + d (B) a + b = c + d (C) $ab=\frac{c+d}{a+b}$ (D) a − b = c − d

Q86. MCQ (Single Correct Answer) · **Mathematics**

If Rolle's theorem holds for the function $f(x)=x^3-ax^2+bx-4, x\in[1,2]$ with $f'\left(\frac{4}{3}\right)=0$, then ordered pair (a, b) is equal to :

Open solution

(A) $(-5,-8)$ (B) $(5,-8)$ (C) $(-5,8)$ (D) $(5,8)$

Q87. Numerical · **Mathematics**

Let $A=\begin{bmatrix}x & y & z\\ y & z & x\\ z & x & y\end{bmatrix}$, where x, y and z are real numbers such that x + y + z > 0 and xyz = 2. If $A^2=I_3$, then the value of $x^3+y^3+z^3$ is _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

Two radioactive substances X and Y originally have N_1 and N_2 nuclei respectively. Half life of X is half of the half life of Y. After three half lives of Y, number of nuclei of both are equal. The ratio $\frac{N_1}{N_2}$ will be equal to :

Open solution

(A) $\frac{1}{8}$ (B) $\frac{3}{1}$ (C) $\frac{1}{3}$ (D) $\frac{8}{1}$

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Q89. MCQ (Single Correct Answer) · **Physics**

Open solution

Given below are two statements :

Statement I : A speech signal of 2 kHz is used to modulate a carrier signal of 1 MHz. The bandwidth requirement for the signal is 4 kHz.

Statement II : The side band frequencies are 1002 kHz and 998 kHz.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

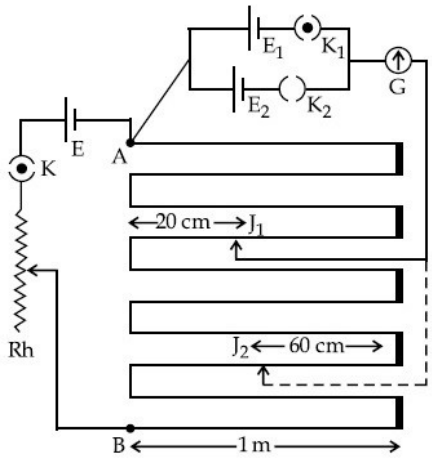
(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Q90. Numerical · **Physics**

Open solution

In the given circuit of potentiometer, the potential difference E across AB (10 m length) is larger than E_1 and E_2 as well. For key K_1 (closed), the jockey is adjusted to touch the wire at point J_1 so that there is no deflection in the galvanometer. Now the first battery (E_1) is replaced by second battery (E_2) for working by making K_1 open and K_2 closed. The galvanometer gives then null deflection at J_2. The value of $\frac{E_1}{E_2}$ is $\frac{a}{b}$, where a = _____.



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Answer Key

Chemistry

1. **A**	2. **B**	3. **D**	4. **D**	5. **D**
6. **C**	7. **A**	8. **C**	9. **B**	10. **D**
11. **B**	12. **A**	13. **A**	14. **D**	15. **4**
16. **741**	17. **5576**	18. **4**	19. **173**	20. **7**
21. **375**	22. **1**	23. **526**		

Mathematics

24. **D**	25. **C**	26. **D**	27. **C**	28. **D**
29. **B**	30. **A**	31. **A**	32. **B**	33. **D**
34. **C**	35. **B**	36. **D**	37. **2**	38. **64**
39. **32**	40. **13**	41. **9**	42. **21**	43. **2**
44. **12**	45. **144**			

Physics

46. **A**	47. **C**	48. **B**	49. **B**	50. **D**
51. **A**	52. **A**	53. **A**	54. **D**	55. **C**
56. **C**	57. **B**	58. **C**	59. **C**	60. **D**
61. **B**	62. **C**	63. **C**	64. **5**	65. **3600**
66. **15**	67. **144**	68. **1**	69. **1**	70. **50**
71. **128**	72. **10**			

Out of syllabus

73. **C**	74. **B**	75. **D**	76. **D**	77. **B**
78. **B**	79. **70**	80. **D**	81. **C**	82. **B**
83. **D**	84. **A**	85. **D**	86. **D**	87. **7**
88. **D**	89. **D**	90. **1**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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